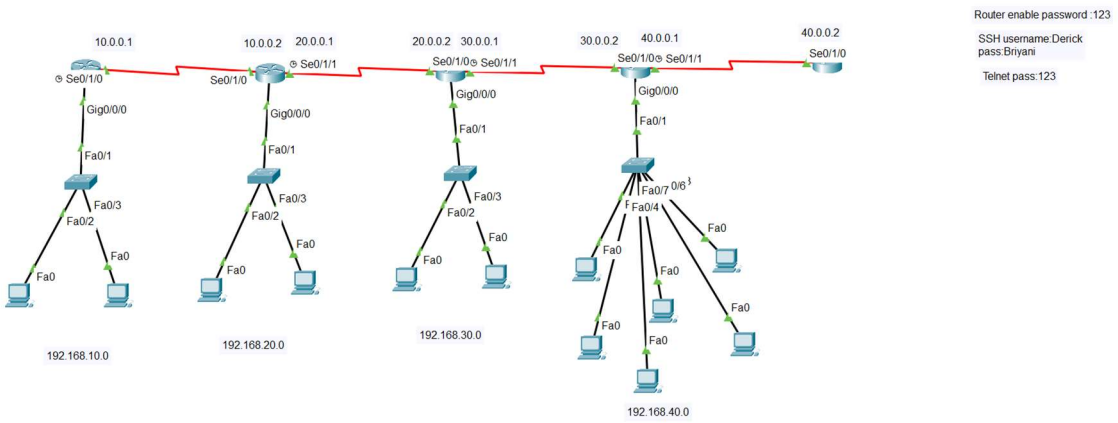
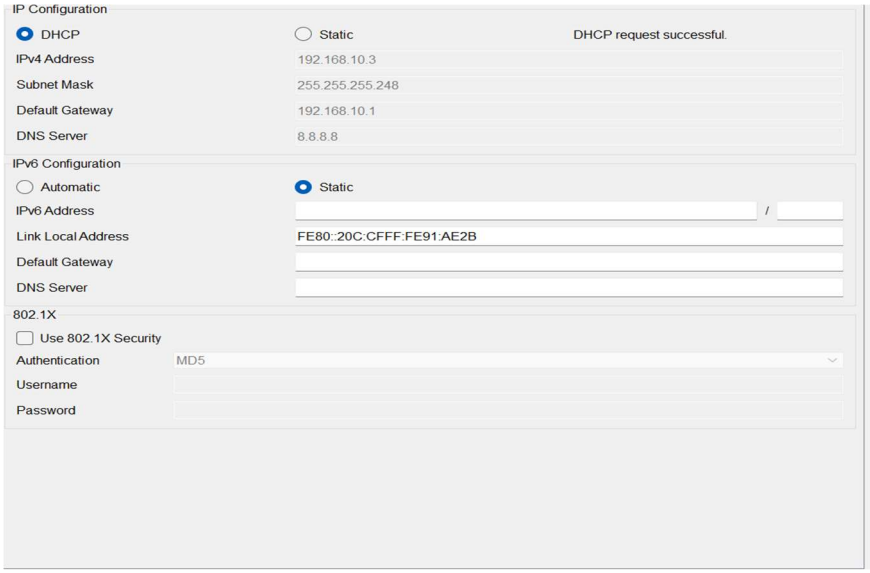
Topology:



DHCP Verification:



Routing verification:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	Switch1	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	Switch2	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC3	Switch0	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC5	Switch1	ICMP		0.000	N	3	(edit)	(delete)

Ping /connectivity testing:

```
Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time=11ms TTL=254
Reply from 192.168.20.1: bytes=32 time=16ms TTL=254
Reply from 192.168.20.1: bytes=32 time=1ms TTL=254
Reply from 192.168.20.1: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 16ms, Average = 7ms

C:\>
```

SSH:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -L Derick 192.168.10.1

Password:
% Login invalid


Password:



R1>
```

Telnet:

```
C:\>telnet 192.168.20.2
Trying 192.168.20.2 ...Open


User Access Verification

Password:
Switch>
```

DNAT:

```
R4(config)#do show ip nat translation
Pro  Inside global     Inside local       Outside local      Outside global
icmp 20.0.0.1:22       192.168.40.7:22    40.0.0.2:22        40.0.0.2:22
icmp 20.0.0.1:23       192.168.40.7:23    40.0.0.2:23        40.0.0.2:23
icmp 20.0.0.1:24       192.168.40.7:24    40.0.0.2:24        40.0.0.2:24
icmp 20.0.0.1:25       192.168.40.7:25    40.0.0.2:25        40.0.0.2:25
icmp 20.0.0.2:5        192.168.40.3:5     40.0.0.2:5         40.0.0.2:5
icmp 20.0.0.2:6        192.168.40.3:6     40.0.0.2:6         40.0.0.2:6
icmp 20.0.0.2:7        192.168.40.3:7     40.0.0.2:7         40.0.0.2:7
icmp 20.0.0.2:8        192.168.40.3:8     40.0.0.2:8         40.0.0.2:8
```